

Chapter 4: Sequences and Series

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4.1 Limits of Sequences

definition using first principle

Reference: Stewart's Calculus 8E, Section 11.1

Goal: Understand the definition of a sequence and its limit, and find the limit using the first principle.

Definition 1. (Sequence)

A **sequence** can be regarded as a list of numbers written **in a definite order**:

$$a_1, a_2, a_3, \dots, a_n, a_{n+1}, \dots$$

The number

- a_1 is called the **first term**,
- a_2 is the **second term**, and
- a_n is the n th term.

We will deal exclusively with infinite sequences, and so each term a_n will have a successor a_{n+1} .

NOTE. The sequence $\{a_1, a_2, a_3, \dots\}$ is also denoted by

$$\{a_n\} \quad \text{or} \quad \{a_n\}_{n=1}^{\infty}.$$

The first term of a sequence does not necessarily have index 1, i.e. n **doesn't have to start with 1**. Indeed, it can start with any integer. For example, we can also represent a sequence by

$$\{a_n\}_{n=0}^{\infty} \quad \text{or} \quad \{a_n\}_{n=2}^{\infty}.$$

Example 4.1. Some sequences can be defined by giving a formula for a_n .

$$(a) \left\{ \frac{n}{n+1} \right\}_{n=1}^{\infty}, \quad a_n = \frac{n}{n+1}.$$

$$\left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots, \frac{n}{n+1}, \dots \right\}.$$

$$(b) \left\{ \cos \frac{n\pi}{6} \right\}_{n=0}^{\infty}, \quad a_n = \cos \frac{n\pi}{6}$$

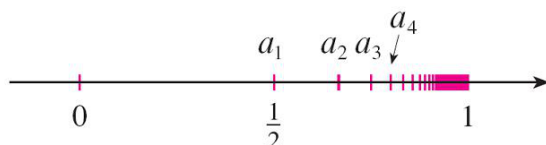
$$\left\{ 1, \frac{\sqrt{3}}{2}, \frac{1}{2}, 0, \dots, \cos \frac{n\pi}{6}, \dots \right\}.$$

$$(c) \left\{ \sqrt{n-3} \right\}_{n=3}^{\infty}, \quad a_n = \sqrt{n-3}$$

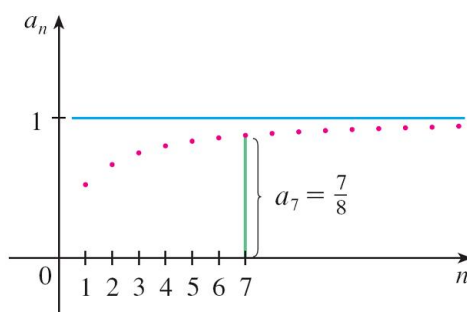
starting index
need to be
at least 3 for
square root to be defined.

$$\{0, 1, \sqrt{2}, \sqrt{3}, \dots, \sqrt{n-3}, \dots\}.$$

A sequence such as $\left\{ \frac{n}{n+1} \right\}_{n=1}^{\infty}$ can be visualised either by plotting its term on number line



or by plotting its graph as in the following figure



* n must be integer so we only have all the dots and not a continuous curve.

In the example above, it seems that the terms $a_n = \frac{n}{n+1}$ in the sequence are approaching 1. We indicate this by writing

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1.$$

In general, the notation

$$\lim_{n \rightarrow \infty} a_n = L$$

means that terms of the sequence $\{a_n\}$ approach L as n gets larger and larger.

$\downarrow$
a real number

Definition 2. (Limit of a Sequence (Informal))

A sequence $\{a_n\}$ has the limit L (a real number) and we write

$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or} \quad a_n \rightarrow L \quad \text{as} \quad n \rightarrow \infty$$

if we can make the terms a_n as close to L as we like by taking n sufficiently large.

If $\lim_{n \rightarrow \infty} a_n$ exists, we say that the sequence $\{a_n\}$ **converges** (or is **convergent**).

Otherwise, we say that the sequence **diverges** (or is **divergent**).

Figures below show the graphs of two sequences that have the limit L .



A more precise definition of limit of a sequence is given as follows.

Definition 3. (Limit of a Sequence (Formal))

A sequence $\{a_n\}$ has the limit L (a real number), and we write

$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or} \quad a_n \rightarrow L \quad \text{as} \quad n \rightarrow \infty$$

$\rightarrow$ dist between a_n and L .

we want a_n to be as close as possible to L : $|a_n - L|$

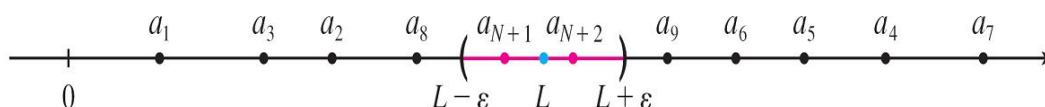
$\hookrightarrow$ we want $|a_n - L| < \varepsilon$ for any $\varepsilon > 0$.

if for every $\varepsilon > 0$ there is a corresponding integer N such that

$$n > N \implies |a_n - L| < \varepsilon.$$

↓
threshold

This definition can be illustrated by the following figure, in which the terms a_n are approaching the limit L .



We can think of $|a_n - L|$ as the distance between a_n and L . The definition says we can make this distance arbitrarily small (smaller than ε for any ε), by making n large enough, say $n > N$ for some integer N .

Example 4.2. Using the formal definition of limit, show that the sequence $\left\{\frac{1}{n}\right\}$ converges to 0, i.e. $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. Goal: For any $\varepsilon > 0$, there exists N s.t. $n > N \implies \left|\frac{1}{n} - 0\right| < \varepsilon$

Solution. Let $\varepsilon > 0$. We want to find a corresponding integer N such that

$$n > N \implies \left|\frac{1}{n} - 0\right| < \varepsilon.$$

We work backwards: From the inequality $\left|\frac{1}{n} - 0\right| < \varepsilon$, we back out a choice of N as follows.

Note that

$$\begin{aligned} \left|\frac{1}{n} - 0\right| < \varepsilon &\iff \frac{1}{n} < \varepsilon \\ &\iff n > \frac{1}{\varepsilon} \end{aligned} \tag{4.1}$$

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$$\lceil 3.17 \rceil = 4$$

$$n > \lceil 3.17 \rceil \Rightarrow n > 3.17 \quad 6$$

Choose $N = \left\lceil \frac{1}{\varepsilon} \right\rceil$ (here $\lceil x \rceil$ denotes the smallest integer larger or equal to x). Then it follows from (4.1) that

$$\begin{aligned} n > N = \left\lceil \frac{1}{\varepsilon} \right\rceil &\Rightarrow n > \frac{1}{\varepsilon} \quad (\text{since } \lceil x \rceil \geq x \text{ for all } x) \\ &\Rightarrow \left| \frac{1}{n} - 0 \right| < \varepsilon, \end{aligned} \quad (4.2)$$

as required.

NOTE: In fact, any integer larger than $1/\varepsilon$ can be chosen as N in the above solution.

For example, we could choose $N = 2 \cdot \left\lceil \frac{1}{\varepsilon} \right\rceil$, or $N = 10 \cdot \left\lceil \frac{1}{\varepsilon} \right\rceil$ etc.

□

Example 4.3. Using the formal definition of limit, show that the sequence $\{a_n\}$,

where $a_n = \frac{n+4}{n+1}$, converges to 1, i.e. $\lim_{n \rightarrow \infty} \frac{n+4}{n+1} = 1$.

SL

Solution. Let $\varepsilon > 0$. The definition requires us to find a number N such that

$$n > N \Rightarrow \left| \frac{n+4}{n+1} - 1 \right| < \varepsilon. \quad (4.3)$$

We have

$$\left| \frac{n+4}{n+1} - 1 \right| = \frac{3}{n+1}.$$

Therefore,

$$\left| \frac{n+4}{n+1} - 1 \right| < \varepsilon \iff \frac{3}{n+1} < \varepsilon \iff n > \frac{3}{\varepsilon} - 1.$$

assume $n \geq 1$

no guarantee that $\frac{3}{\varepsilon} - 1$ is positive!

$$\frac{3}{n} < \varepsilon \iff n > \frac{3}{\varepsilon} \iff n > \left\lceil \frac{3}{\varepsilon} \right\rceil. \text{ Choose } N = \left\lceil \frac{3}{\varepsilon} \right\rceil$$

Hence, we can choose $N = \left\lceil \frac{3}{\varepsilon} - 1 \right\rceil$ for (4.3) to be valid.

□

prof prefer this way

$$n > N \Rightarrow \left| \frac{n+4}{n+1} - 1 \right| < \varepsilon$$

Extra Example

Use definition to show.

$$\lim_{n \rightarrow \infty} \frac{100-n-\sqrt{n}}{n+1} = -1$$

Want: For $\varepsilon > 0$, $\exists N$ s.t

$$n > N \Rightarrow \left| \frac{100-n-\sqrt{n}}{n+1} - (-1) \right| < \varepsilon$$

$$\left| \frac{100-n-\sqrt{n}}{n+1} + 1 \right| < \varepsilon$$

$$\iff \left| \frac{101-\sqrt{n}}{n+1} \right| < \varepsilon$$

$$\iff \frac{\sqrt{n}-101}{n+1} < \varepsilon$$

$$\text{if } n > 101^2$$

threshold is max of these 2 numbers

$$\iff \frac{\sqrt{n}-101}{n} < \varepsilon$$

$$\iff \frac{\sqrt{n}}{n} < \varepsilon$$

$$\iff n > \left(\frac{1}{\varepsilon}\right)^2$$

$$\text{let } N = \max \left\{ 101^2, \left(\frac{1}{\varepsilon}\right)^2 \right\}$$

Then for $n > N$, we can reverse all the inequalities above and arrive at

$$\left| \frac{100-n-\sqrt{n}}{n+1} - (-1) \right| < \varepsilon$$

Definition 4. (Subsequence)

Given a sequence $\{a_n\}_{n=1}^{\infty}$, a **subsequence** is a sequence of the form

$$a_{n_1}, a_{n_2}, a_{n_3}, \dots,$$

where

$$1 \leq n_1 < n_2 < n_3 < \dots$$

Note: $n_k \geq k$ for all $k = 1, 2, 3, \dots$

For example,

- $\{a_1, a_3, a_5, \dots\}$ is a subsequence of $\{a_n\}_{n=1}^{\infty}$. This is called the **odd subsequence** since it consists of all the odd terms (terms with odd indices).
- $\{a_2, a_4, a_6, \dots\}$ is another subsequence of $\{a_n\}_{n=1}^{\infty}$. This is called the **even subsequence** since it consists of all the even terms (terms with even indices).

e.g. of subsequence:

$$a_2, a_4, a_7, a_8, a_{100}, a_{102}, a_{105}, \dots$$

$$1 \leq n_1 = 2 < n_2 = 4 < n_3 = 7 < n_4 = 8 < n_5 = 100, \dots$$

$$i < j, n_i < n_j$$

$$n_k \geq k \text{ for all } k = 1, 2, 3, \dots$$

The following result is useful when determining the convergence of a sequence using its subsequences.

Theorem 1. (Subsequence Test)

If a sequence $\{a_n\}$ converges to L , then every subsequence of $\{a_n\}$ converges to L .

Proof. Let $\varepsilon > 0$. Since $\{a_n\}$ converges to L , there exists a positive integer N such that

$$n > N \implies |a_n - L| < \varepsilon. \quad (4.4)$$

Consider a subsequence $\{b_k\}$ of $\{a_n\}$, where

$$b_k = a_{n_k}, \quad k = 1, 2, \dots$$

Note that for all $k > N$, we have $n_k \geq k > N$. So by (4.4),

$$|b_k - L| = |a_{n_k} - L| < \varepsilon.$$

Thus, we have proved that

$$k > N \implies |b_k - L| < \varepsilon,$$

In view of definition of limit. Let $\varepsilon > 0$.
WTF threshold $K(\varepsilon)$, s.t.
 $k > K \implies |b_k - L| < \varepsilon$
(where $b_k = a_{n_k}$)

Choose $K = K(\varepsilon) = N$
see above for the condition that N represents

$$k > K \implies n_k > n_K \geq K = N$$

$$\implies |a_{n_k} - L| < \varepsilon$$

$$\text{i.e. } |b_k - L| < \varepsilon$$

as required.

$$\text{WTS } a_{n_k} \rightarrow L$$

that is, the subsequence $\{b_k\}$ also converges to L . □

Here is how we can use the Subsequence Test to show **divergence**: If $a_n \rightarrow L$, then every $a_{n_k} \rightarrow L$

There exist two subsequences of $\{a_n\}$ that converge to different limits	$\implies \{a_n\}$ diverges .
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There exists one subsequence of $\{a_n\}$ that diverges	$\implies \{a_n\}$ diverges .
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Example 4.4. Use the Subsequence Test to show that the sequence $\{(-1)^n\}$ is divergent.

Solution. Note that

- the even subsequence is $-1, -1, -1, \dots$ which converges to -1
- the odd subsequence is $1, 1, 1, \dots$ which converges to 1 .

Since these two subsequence converge to different limits, we conclude by the Subsequence Test (Theorem 1) that the limit of the sequence $\{(-1)^n\}$ does not exist. □

Example 4.5. Consider the sequence $\{a_n\}_{n=1}^{\infty}$ where a_n is defined by

$$a_n = \begin{cases} m & \text{if } n = 2^m \text{ for some } m \\ 0 & \text{otherwise.} \end{cases}$$

Verify that this sequence is divergent.

Solution. Consider the subsequence $\{b_m\}_{m=1}^{\infty}$, where $b_m = a_{2^m}$. We have

$$b_1 = 1, b_2 = 2, \dots, b_m = m.$$

Clearly, $\lim_{m \rightarrow \infty} b_m = \lim_{m \rightarrow \infty} m = \infty$, i.e. the sequence $\{b_m\}$ is divergent. By the Subsequence Theorem, the sequence $\{a_n\}$ is divergent. □

Example 4.5
Consider the sequence $\{a_n\}_{n=1}^{\infty}$ where a_n is defined by

$$a_n = \begin{cases} m & \text{if } n = 2^m \text{ for some } m \\ 0 & \text{otherwise.} \end{cases}$$

 Verify that this sequence is divergent.
 Consider the subsequence given by

$$\{a_{2^m}\}_{m=1}^{\infty} = \{0, 1, 2, 3, 4, \dots\}$$

$\lim_{m \rightarrow \infty} a_{2^m} = \lim_{m \rightarrow \infty} m = \infty$
 $\implies \{a_{2^m}\}$ diverges
 $\downarrow$
 $\therefore$ By subsequence test, $\{a_n\}$ diverges

Divergent sequences can behave in many unpredictable ways. There is a class of divergent sequences that diverge to ∞ .

Examples:

$$\lim_{n \rightarrow \infty} n = \infty, \quad \lim_{n \rightarrow \infty} \ln n = \infty.$$

We say that the sequence $\{n\}$, $\{\ln n\}$ **diverges** to infinity.

Some useful properties:



Suppose $a_n > 0$.

$$a_n \rightarrow 0 \implies \frac{1}{a_n} \rightarrow \infty$$

$$a_n \rightarrow \infty \implies \frac{1}{a_n} \rightarrow 0.$$

$$b_n \geq a_n \text{ and } a_n \rightarrow \infty \implies b_n \rightarrow \infty.$$

Example 4.6. Find the limit

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt{10+n}}.$$

Solution. Divide both the numerator and denominator by n :

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{n}{\sqrt{10+n}} &= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{\frac{10}{n^2} + \frac{1}{n}}} \\ &= \infty, \end{aligned}$$

since

$$\sqrt{\frac{10}{n^2} + \frac{1}{n}} \rightarrow 0.$$

□

Example 4.7. Find the limit of $\frac{(n!)^{1/n}}{n^{0.1}}$ as $n \rightarrow \infty$.

Note that

$$\begin{aligned}
 n! &= n(n-1) \cdots \left(\left\lfloor \frac{n}{2} \right\rfloor + 1 \right) \left(\left\lfloor \frac{n}{2} \right\rfloor \right) \cdots 2 \cdot 1 \\
 &\geq n(n-1) \cdots \left(\left\lfloor \frac{n}{2} \right\rfloor + 1 \right) \\
 &\geq \underbrace{\frac{n}{2} \frac{n}{2} \cdots \frac{n}{2}}_{n - \left\lfloor \frac{n}{2} \right\rfloor \text{ times}} \\
 &\geq \left(\frac{n}{2} \right)^{n/2}.
 \end{aligned}$$

Handwritten notes:
 no. of terms = $n - \left\lfloor \frac{n}{2} \right\rfloor$
 no. of terms = $\left\lfloor \frac{n}{2} \right\rfloor$
 need floor function as n may be odd number
 no. of terms $\geq \frac{n}{2}$

Taking n -th root on both sides, we have

$$(n!)^{1/n} \geq \left(\frac{n}{2} \right)^{1/2}.$$

Dividing both sides by $n^{0.1}$, we have

$$\frac{(n!)^{1/n}}{n^{0.1}} \geq \frac{n^{0.4}}{\sqrt{2}}.$$

Since $\lim_{n \rightarrow \infty} \frac{n^{0.4}}{\sqrt{2}} = \infty$, we deduce that

$$\lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n^{0.1}} = \infty.$$

□

4.2 Finding Limits of Sequences

Reference: Stewart's Calculus 8E, Section 11.1

Goal: Find the limit of a sequence using Limit Laws, the Squeeze Theorem, functions defining sequences, and the continuity of a function of convergent sequences.

Sometimes, a complicated sequence can be broken down into smaller (easier) sequences by means of addition/subtraction or multiplication/division. The following limit properties allow us to compute limit of a sequence from its smaller constituents.

Limit Laws for Sequences.

Suppose $\{a_n\}$ and $\{b_n\}$ are convergent sequences (this assumption is important), and c is a constant. Then

$$(a) \lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

$$(b) \lim_{n \rightarrow \infty} (a_n - b_n) = \lim_{n \rightarrow \infty} a_n - \lim_{n \rightarrow \infty} b_n$$

$$(c) \lim_{n \rightarrow \infty} c \cdot a_n = c \cdot \lim_{n \rightarrow \infty} a_n \quad \text{factor out constant}$$

$$(d) \lim_{n \rightarrow \infty} a_n b_n = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$$

$$(e) \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}, \quad \text{provided } \lim_{n \rightarrow \infty} b_n \neq 0.$$

$$(f) \lim_{n \rightarrow \infty} (a_n)^p = \left(\lim_{n \rightarrow \infty} a_n \right)^p, \quad \text{provided } p > 0 \text{ and } a_n > 0. \quad \text{factor out power}$$

Example 4.8. Find $\lim_{n \rightarrow \infty} \frac{n}{n+1}$.

Solution. Instead of evaluating this from scratch using the definition, we can make use of the Limit Laws.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{n}{n+1} &= \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} \\ &= \frac{\lim_{n \rightarrow \infty} 1}{\lim_{n \rightarrow \infty} 1 + \lim_{n \rightarrow \infty} \frac{1}{n}} \quad (\text{by Limit Law}) \\ &= \frac{1}{1+0} = 1. \end{aligned}$$

since $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$. □

Example 4.9. Find the limit

$$\lim_{n \rightarrow \infty} \frac{3n+20}{\sqrt{n^2+n+\cos n\pi}}. \quad \text{as } n \rightarrow \infty, \cos n\pi = (-1)^n$$

Solution.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{3n+20}{\sqrt{n^2+n+\cos n\pi}} &= \lim_{n \rightarrow \infty} \frac{3 + \frac{20}{n}}{\frac{1}{n} \sqrt{n^2+n+\cos n\pi}} \\ &= \lim_{n \rightarrow \infty} \frac{3 + \frac{20}{n}}{\sqrt{\frac{n^2+n+\cos(n\pi)}{n^2}}} \\ &= \lim_{n \rightarrow \infty} \frac{3 + \frac{20}{n}}{\sqrt{1 + \frac{1}{n} + \frac{\cos(n\pi)}{n^2}}} \end{aligned}$$

Since $\cos(n\pi)$ is bounded, $\frac{\cos(n\pi)}{n^2} \rightarrow 0$ (we can show this by Squeeze if we want).

Hence, by the Limit Laws,

$$\lim_{n \rightarrow \infty} \frac{3n+20}{\sqrt{n^2+n+\cos n\pi}} = \frac{3+0}{\sqrt{1+0+0}} = 3.$$

□

In addition to the Limit Laws, the following results are useful in finding the limits of sequences. Their proofs are omitted.

Theorem 2. (Sequence defined by a Function)

If $\lim_{x \rightarrow \infty} f(x) = L$ and $f(n) = a_n$ when n is an integer, then

$$\lim_{n \rightarrow \infty} a_n = L.$$

pass to function

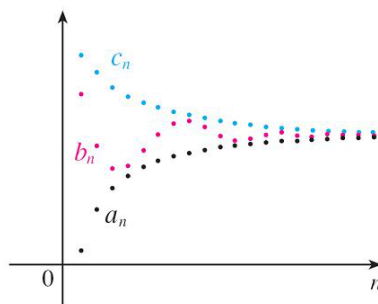
$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} f(n) = \lim_{x \rightarrow \infty} f(x)$$

Theorem 3. (Squeeze Theorem for Sequences) *Squeeze Theorem*

If $a_n \leq b_n \leq c_n$ for $n \geq N$, and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then

$$\lim_{n \rightarrow \infty} b_n = L.$$

The following figure illustrates the sequence b_n being squeezed between a_n and c_n when n becomes large.



The following theorem says that if we apply a continuous function to the terms of a convergent sequence, the result is also convergent.

Theorem 4. (Continuous Function for Convergent Sequences)

If $\lim_{n \rightarrow \infty} a_n = L$ and f is a **continuous** function at L , then

$$\lim_{n \rightarrow \infty} f(a_n) = f(\lim_{n \rightarrow \infty} a_n) = f(L).$$

$$\lim_{n \rightarrow \infty} \sqrt{a_n} = \sqrt{\lim_{n \rightarrow \infty} a_n}$$

$$\lim_{n \rightarrow \infty} \sin(a_n) = \sin(\lim_{n \rightarrow \infty} a_n)$$

$$\lim_{n \rightarrow \infty} e^{a_n} = e^{\lim_{n \rightarrow \infty} a_n}$$

Example 4.10. Prove that if $\lim_{n \rightarrow \infty} |a_n| = 0$ then $\lim_{n \rightarrow \infty} a_n = 0$.

Solution. Notice

$$-|a_n| \leq a_n \leq |a_n|$$

for all n . Since $\lim_{n \rightarrow \infty} -|a_n| = \lim_{n \rightarrow \infty} |a_n| = 0$, we deduce from the Squeeze theorem that

$$\lim_{n \rightarrow \infty} a_n = 0.$$

□

Example 4.11. Find $\lim_{n \rightarrow \infty} \sin(\pi/n)$.

Solution. The function $\sin x$ is continuous at 0. By Theorem 4 we have

$$\lim_{n \rightarrow \infty} \sin(\pi/n) = \sin\left(\lim_{n \rightarrow \infty} (\pi/n)\right) = \sin 0 = 0.$$

□

Example 4.12. Determine whether the limit of the sequence $\{a_n\}$, where $a_n = \frac{n!}{n^n}$, exists.

Solution. Notice

$$a_n = \frac{1 \cdot 2 \cdot 3 \cdots n}{n \cdot n \cdot n \cdots n}.$$

Since $\frac{i}{n} \leq 1$ for all $2 \leq i \leq n$, we have

keep $\frac{1}{n}$ to show $\rightarrow 0$

$$0 \leq a_n \leq \frac{1}{n}.$$

Clearly, $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$. By Squeeze theorem,

$$\lim_{n \rightarrow \infty} a_n = 0.$$

□

NOTE: The result says that n^n “grows faster” than $n!$.

In general :

$\log / \ln \ll \text{polynomial} \ll \text{exponential} \ll \text{factorial} \ll n^n$

Example 4.13. Limit of a geometric sequence:

$$\lim_{n \rightarrow \infty} r^n = \begin{cases} 0 & \text{if } -1 < r < 1 \\ 1 & \text{if } r = 1 \\ \infty \text{ (divergent)} & \text{if } r > 1 \\ \text{divergent} & \text{if } r \leq -1. \end{cases}$$

Solution. It is obvious that

$$\lim_{n \rightarrow \infty} 1^n = 1,$$

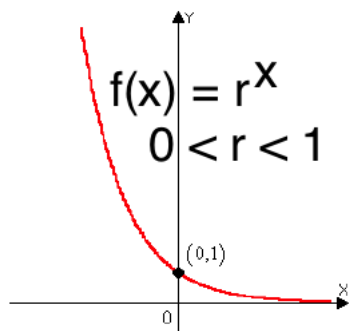
$$\lim_{n \rightarrow \infty} (-1)^n \text{ is divergent}$$

So the result is true for $r = 1, -1$. It remains to consider the case: $-1 < r < 1$, $r > 1$ and $r < -1$.

Case 1: $-1 < r < 1$.

If $r = 0$, then $\lim_{n \rightarrow \infty} r^n = 0$. So we may assume that $r \neq 0$.

Suppose $0 < r < 1$. We apply Theorem 2 (Sequence defined by a Function) to the exponential function $f(x) = r^x$.



Thus,

$$\lim_{n \rightarrow \infty} r^n = \lim_{x \rightarrow \infty} r^x = 0.$$

Suppose $-1 < r < 0$. Then

$$-|r|^n \leq r^n \leq |r|^n.$$

By the preceding paragraph,

$$\lim_{n \rightarrow \infty} |r|^n = 0.$$

By Squeeze, we deduce that

$$\lim_{n \rightarrow \infty} r^n = 0.$$

Case : Alternatively.

Let $\varepsilon > 0$. Working backward

$$|r^n - 0| < \varepsilon$$

$$\Leftrightarrow r^n < \varepsilon$$

$$\Leftrightarrow \ln r^n < \ln \varepsilon$$

$$\Leftrightarrow n \ln r < \ln \varepsilon$$

$$\Leftrightarrow n > \frac{\ln \varepsilon}{\ln r} \quad \because \ln r < 0 \text{ when } 0 < r < 1$$

$$\text{Choose } N = \left\lceil \frac{\ln \varepsilon}{\ln r} \right\rceil$$

Then $n > N$, reverse all steps above & arrive at

$$|r^n - 0| < \varepsilon$$

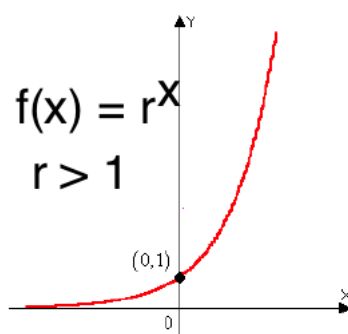
$$\text{i.e. } \lim_{n \rightarrow \infty} r^n = 0$$

Squeeze
Theorem

Just make use of
previous case.

Case 2: $r > 1$.

We apply Theorem 2 (Sequence defined by a Function) to the exponential function $f(x) = r^x$. Since $r > 1$, we have



Alternative: assume for a contradiction that

$$r^n < M \quad \text{for some } M$$

$$\Rightarrow \ln r^n < \ln M$$

$$n \ln r < \ln M$$

$$n < \frac{\ln M}{\ln r} \quad \because \ln r > 0 \text{ for } r > 1$$

Thus,

$$\lim_{n \rightarrow \infty} r^n = \lim_{x \rightarrow \infty} r^x = \infty.$$

Case 3: $r < -1$.

The even subsequence is

$$|r|^2, |r|^4, |r|^6, \dots$$

Since $|r| > 1$, by Case 2, the subsequence diverges to ∞ . Hence, by the Subsequence Test (Theorem 1), the sequence r^n is divergent.

□

When calculating limits of functions, the following **L'Hopital's Rule** can be useful:

Suppose f and g are differentiable and $g'(x) \neq 0$, and

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

or

$$\lim_{x \rightarrow a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \pm\infty$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

indeterminate form
 $\left\{ \frac{f(a)}{g(a)} = \frac{0}{0}, \frac{+\infty}{-\infty} \right\}$
must be a real number, $+\infty$ or $-\infty$

Here, a can be any real number or $\pm\infty$.

Example 4.14. Calculate $\lim_{n \rightarrow \infty} a_n$, where

$$a_n = \frac{n + \ln n}{n^2}.$$

Squeeze theorem also can solve!

$$0 \leq a_n = \frac{n + \ln n}{n^2} \leq \frac{n + n}{n^2} = \frac{2}{n}$$

Since $\frac{2}{n} \rightarrow 0$ for $n \geq 1$, by squeeze, $a_n \rightarrow 0$

Solution. The sequence is defined by the function $f(x) = \frac{x + \ln x}{x^2}$. By Theorem 2 (Sequence defined by a Function), the limit can be evaluated using L'Hopital's Rule as follows:

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{x \rightarrow \infty} f(x) \\ &= \lim_{x \rightarrow \infty} \frac{x + \ln x}{x^2} \quad \begin{matrix} \rightarrow \infty \\ \rightarrow \infty \end{matrix} \\ &= \lim_{x \rightarrow \infty} \frac{1 + (1/x)}{2x} \\ &= \lim_{x \rightarrow \infty} \frac{1}{2x} + \frac{1}{2x^2} \\ &= 0 + 0 = 0. \end{aligned}$$

□

Example 4.15. Calculate $\lim_{n \rightarrow \infty} e^{3n/(n+1)}$.

Solution. We have $e^{3n/(n+1)} = f(a_n)$, where

$$a_n = \frac{3n}{n+1}, \quad f(x) = e^x.$$

CHAPTER 4: SEQUENCES AND SERIES

Furthermore,

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{3n}{n+1} = \lim_{n \rightarrow \infty} \frac{3}{1 + 1/n} = 3.$$

By Theorem 2 (Sequence defined by a Function) and Theorem 4 (Continuous Function for Convergent Sequence),

$$\begin{aligned} \lim_{n \rightarrow \infty} e^{3n/(n+1)} &= \lim_{n \rightarrow \infty} f(a_n) \\ &= f\left(\lim_{n \rightarrow \infty} a_n\right) \\ &= f(3) \\ &= e^3. \end{aligned}$$

Bonus

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sin x - x}{\ln x} &= \lim_{x \rightarrow \infty} \frac{\sin x - x}{\ln x} \\ &= \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(\sin x - x)}{\frac{d}{dx} \ln x} \quad (\text{L'Hopital}) \\ &= \lim_{x \rightarrow \infty} \frac{\cos x - 1}{\frac{1}{x}} \rightarrow \text{not real! should not have used} \\ &= \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(\cos x - 1)}{\frac{d}{dx} \frac{1}{x}} \quad (\text{L'Hopital}) \\ &= \lim_{x \rightarrow \infty} \frac{-\sin x}{-\frac{1}{x^2}} \\ &= \lim_{x \rightarrow \infty} x^2 \sin x = \pm \infty \rightarrow \text{mathematical rule wrong!} \end{aligned}$$

Shouldn't have used L'Hopital cause it doesn't have the indeterminate form.

OR: $\frac{\sin n - n}{\ln n} \leq \frac{1 - n}{\ln n}$ $\because \sin n \leq 1$
 $\lim_{n \rightarrow \infty} \frac{1 - n}{\ln n} = \lim_{x \rightarrow \infty} \frac{1 - x}{\ln x} = \lim_{x \rightarrow \infty} \frac{-1}{\frac{1}{x}} = \lim_{x \rightarrow \infty} (-x) = -\infty$

Example 4.16. Find the limit

$$\lim_{n \rightarrow \infty} (\sin^2 3n) 3^{n+1} 7^{-n}.$$

$$\begin{aligned} -1 &\leq \sin x \leq 1 \\ 0 &\leq \sin^2 x \leq 1 \end{aligned}$$

Solution. We apply Squeeze Theorem. Note that $0 \leq \sin^2 3n \leq 1$. So

$$0 \leq (\sin^2 3n) 3^{n+1} 7^{-n} \leq 3^{n+1} 7^{-n} = 3 \left(\frac{3}{7}\right)^n.$$

Note that $\lim_{n \rightarrow \infty} \left(\frac{3}{7}\right)^n = 0$ (this is geometric sequence that converges to 0). So $\lim_{n \rightarrow \infty} 3 \left(\frac{3}{7}\right)^n = 3 \lim_{n \rightarrow \infty} \left(\frac{3}{7}\right)^n = 3 \cdot 0 = 0$.

By Squeeze Theorem,

$$\lim_{n \rightarrow \infty} (\sin^2 3n) 3^{n+1} 7^{-n} \leq 3^{n+1} 7^{-n} = 0.$$

4.3 Monotonic Sequences

Reference: Stewart's Calculus 8E, Section 11.1

Goal: Use the Monotonic Sequence Theorem to find the limit of a sequence defined recursively.

Definition 5. (Monotonic Sequences)

A sequence $\{a_n\}_{n=1}^{\infty}$ is called **increasing** if $a_n \leq a_{n+1}$ for all $n \geq 1$.

It is called **decreasing** if $a_n \geq a_{n+1}$ for all $n \geq 1$.

A sequence is **monotonic** if it is increasing or decreasing. $\rightarrow \{5, 5, 5, 5, \dots\}$

If all the inequalities above are **strict**, we say that the sequence is strictly increasing or strictly decreasing respectively. $\Rightarrow a_n > a_{n+1}$ or $a_n < a_{n+1}$

Example 4.17. Show that the sequence $\{a_n\}_{n=1}^{\infty}$, where $a_n = \frac{n}{n^2 + 1}$, is strictly decreasing.

! work backwards.

Solution. We must show that $a_n > a_{n+1}$ for all $n \geq 1$. Indeed,

$$\begin{aligned} a_n > a_{n+1} &\iff \frac{n}{n^2 + 1} > \frac{n+1}{(n+1)^2 + 1} \\ &\iff n((n+1)^2 + 1) > (n+1)(n^2 + 1) \\ &\iff n^3 + 2n^2 + 2n > n^3 + n^2 + n + 1 \\ &\iff n^2 + n > 1, \end{aligned}$$

which is true for all $n \geq 1$.

Alternatively:
consider $f(x) = \frac{x}{x^2 + 1}$
 $f'(x) = \frac{(x^2 + 1) - x(2x)}{(x^2 + 1)^2}$
 $= \frac{1 - x^2}{(x^2 + 1)^2}$ *no equal for the sign*
 $\Rightarrow \left\{ \frac{n}{n^2 + 1} \right\}$ is strictly decreasing for $n > 1$

Definition 6. (Boundedness)

A sequence $\{a_n\}_{n=1}^{\infty}$ is **bounded above** if there is a number M such that

$$a_n \leq M \text{ for all } n \geq 1.$$

It is **bounded below** if there is a number m such that

$$m \leq a_n \text{ for all } n \geq 1.$$

If it is bounded **above and below**, then $\{a_n\}$ is called a **bounded sequence**.

An important result about bounded sequence is the following theorem, whose proof is omitted.

Theorem 5. (Monotonic Sequence Theorem)
 $\left. \begin{array}{l} \text{bounded above + increasing} \Rightarrow \text{convergent} \\ \text{bounded below + decreasing} \Rightarrow \text{convergent} \end{array} \right\}$

 Every bounded, monotonic sequence is convergent.

Example 4.18. Show that the following sequence is bounded and increasing:

$$a_0 = 0, \quad a_1 = \sqrt{2}, \quad a_2 = \sqrt{2 + \sqrt{2}}, \quad a_3 = \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \quad \dots$$

Hence, prove that $\lim_{n \rightarrow \infty} a_n$ exists and compute its value.

Solution. The sequence is defined recursively by

$$a_0 = 0, \quad a_{n+1} = \sqrt{2 + a_n}.$$

Step 1: Show that $\{a_n\}$ is bounded above by 2.

Clearly, $a_0 < 2$. Then $a_1 = \sqrt{2 + a_0} < \sqrt{2 + 2} = 2$. Thus, $a_0 < 2$ implies that $a_1 < 2$.

Now, $a_2 = \sqrt{2 + a_1} < \sqrt{2 + 2} = 2$. So $a_1 < 2$ implies that $a_2 < 2$. In general, $a_n < 2$ implies that $a_{n+1} < 2$.

Subsequently, $a_n < 2$ for all n (formally speaking, this is proof by induction on n).

Step 2: Show that $\{a_n\}$ is increasing. $\Rightarrow a_{n+1} \geq a_n$

prove by induction $\leftarrow$

Since a_n is positive and $a_n < 2$, we have

$$a_{n+1} = \sqrt{2 + a_n} > \sqrt{a_n + a_n} = \sqrt{2a_n} > \sqrt{a_n a_n} = a_n.$$

$\downarrow$ $\text{cause } 2 > a_n$ $\downarrow$ $\text{cause } 2 > a_n$ $\because a_n \text{ is non negative}$

Thus, $a_{n+1} > a_n$ for all $n \geq 0$, and so $\{a_n\}$ is increasing.

Step 3: Find the limit. By Monotonic Sequence Theorem, we know that

$$\lim_{n \rightarrow \infty} a_n = L$$

for some real number L .

We make use of the recurrence to find the limit. First, note that

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} a_n = L.$$

Therefore,

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} a_{n+1} \\ &= \lim_{n \rightarrow \infty} \sqrt{2 + a_n} \\ &= \sqrt{\lim_{n \rightarrow \infty} (2 + a_n)} \quad (\text{by Theorem } \boxed{4}) \\ &= \sqrt{2 + L} \end{aligned}$$

Hence,

$$\begin{aligned} L = \sqrt{2 + L} &\implies L^2 - L - 2 = 0 \\ (L - 2)(L + 1) &= 0, \end{aligned}$$

whence

$$L = 2, \text{ or } L = -1.$$

Since $L \geq 0$, the case $L = -1$ is not possible. We conclude that $L = 2$. □

$\downarrow$
 $\because$ all terms in sequence are negative

4.4 Series

Reference: Stewart's Calculus 8E, Section 11.2

Goal: Understand the meaning and basic properties of a convergent series.

If we add the terms of an infinite sequence $\{a_n\}_{n=1}^{\infty}$, we get an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots,$$

which is called an **infinite series** (or just **series**), and is denoted by

$$\sum_{n=1}^{\infty} a_n \quad \text{or} \quad \sum a_n.$$

QUESTION: We know how to sum finitely many terms. But what do we mean by a sum of infinitely many terms?

ANSWER. For example, the series

$$1 + 2 + 3 + 4 + \cdots + n + \cdots$$

does not seem to converge because if we start adding the terms we get cumulative sums:

$$\begin{array}{ccccccc} & \nearrow s_1 & \nearrow s_2 & \nearrow s_3 & \nearrow s_4 & & \\ a_1 & a_1 + a_2 & a_1 + a_2 + a_3 & a_1 + a_2 + a_3 + a_4 & & & \\ \uparrow & \uparrow & \uparrow & \uparrow & & & \\ 1, & 1 + 2 = 3, & 1 + 2 + 3 = 6, & 1 + 2 + 3 + 4 = 10, & 1 + 2 + 3 + 4 + 5 = 15, & \dots \end{array}$$

which becomes very large (going to ∞) as n increases.

$$\{s_1, s_2, s_3, s_4, \dots, s_k\} = \{1, 3, 6, 10, \dots\} = \left\{ \frac{k(k+1)}{2} \right\}_{k=1}^{\infty}$$

The idea of using cumulative sums (formally called **partial sums**) seems to make sense when speaking about whether a series has a sum. We make this precise in the following definition.

Definition 7. (Series)

Given a series $\sum_{n=1}^{\infty} a_n$, let s_k denote its k -th **partial sum**:

$$s_k = a_1 + a_2 + \cdots + a_k = \sum_{n=1}^k a_n.$$

- (i) If the sequence $\{s_k\}$ converges to a **real number S** , then the series $\sum a_n$ is called **convergent** and we write

$$\sum_{n=1}^{\infty} a_n = S.$$

The number S is called the **sum of the series**.

- (ii) If the sequence $\{s_k\}$ diverges, then the series $\sum_{n=1}^{\infty} a_n$ is called **divergent**.

In short, the sum of a series is the limit of the sequence of its partial sums:

$$\sum_{n=1}^{\infty} a_n = \lim_{k \rightarrow \infty} s_k = \lim_{k \rightarrow \infty} \underbrace{\sum_{n=1}^k a_n}_{\text{partial sum}}.$$

Example 4.19. What is wrong with the following calculations?

$$\begin{aligned} 0 &= 0 + 0 + 0 + \cdots \\ &= (1 - 1) + (1 - 1) + (1 - 1) + \cdots \\ &= 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + \cdots \\ &= 1 + (-1 + 1) + (-1 + 1) + \cdots \\ &= 1 + 0 + 0 + \cdots \\ &= 1. \end{aligned}$$

$$1 - 1 + 1 - 1 + 1 - 1 + \cdots$$

$$\begin{aligned} s_1 &= 1 \\ s_2 &= 1 - 1 = 0 \\ s_3 &= 1 - 1 + 1 = 1 \\ s_4 &= 1 - 1 + 1 - 1 = 0 \end{aligned} \quad \therefore \{s_k\} = \{1, 0, 1, 0, \dots\} \text{ diverges, so}$$

by definition $1 - 1 + 1 - 1 + \dots$ diverges

→ this series diverges
↓
doesn't make sense to attach a number to it

Example 4.20. An important example of a series is the **geometric series**

$$a + ar + ar^2 + ar^3 + \cdots = \sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=0}^{\infty} ar^n$$

where a is a nonzero constant and r is the **common ratio**.

Find the partial sum s_k of the geometric series.

Solution. There are two cases: $r = 1$ and $r \neq 1$.

- If $r = 1$, then

$$s_k = \underbrace{a + a + \cdots + a}_{k \text{ times}} = ka.$$

- If $r \neq 1$, we have

$$s_k = a + ar + ar^2 + \cdots + ar^{k-1}$$

and

$$rs_k = ar + ar^2 + ar^3 + \cdots + ar^k.$$

Subtracting these equations, we get

$$s_k - rs_k = a - ar^k,$$

whence

$$s_k = \frac{a(1 - r^k)}{1 - r}.$$

□

Theorem 6. (Geometric Series)

The geometric series

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \cdots$$

- is convergent if $|r| < 1$ and its sum is

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \quad |r| < 1.$$

- is divergent if $|r| \geq 1$.

Proof. From the preceding example, the partial sum is $s_k = \frac{a(1-r^k)}{1-r}$ if $r \neq 1$.

Case 1: $|r| < 1$.

Then

$$\begin{aligned} \sum_{n=1}^{\infty} ar^{n-1} &= \lim_{k \rightarrow \infty} s_k \\ &= \lim_{k \rightarrow \infty} \frac{a(1-r^k)}{1-r} \\ &= \frac{a(1-\lim_{k \rightarrow \infty} r^k)}{1-r} \quad (\text{Limit Laws}) \\ &= \frac{a}{1-r} \end{aligned}$$

where the last equality follows from the fact that $\lim_{k \rightarrow \infty} r^k = 0$, for $|r| < 1$ (Example 4.13).

Case 2: $|r| \geq 1$. $\rightarrow r=1$
 $\rightarrow r=-1 \Rightarrow \{r^k\} = \{(-1)^k\}$ diverges
 $\rightarrow |r| > 1$

If $r = 1$, then

$$s_k = ka \rightarrow \begin{cases} +\infty, & a > 0 \\ -\infty, & a < 0 \end{cases}$$

so the series diverges.

If $|r| > 1$, then by Example 4.13, $\lim_{k \rightarrow \infty} r^k$ is divergent.

Hence, the series $\sum_{n=1}^{\infty} ar^{n-1} = \lim_{k \rightarrow \infty} s_k$ is divergent. □

Example 4.21. Is the series

$$\sum_{n=1}^{\infty} 2^{2n} 3^{1-n}$$

convergent or divergent?

Solution. Let's rewrite the n th term of the series in the form ar^{n-1} :

$$2^{2n} 3^{1-n} = (2^2)^n \cdot 3 \cdot (3^{-1})^n = 3 \cdot \left(\frac{4}{3}\right)^n = 3 \left(\frac{4}{3}\right) \left(\frac{4}{3}\right)^{n-1} = 4 \left(\frac{4}{3}\right)^{n-1}.$$

The series is a geometric series

$$\sum_{n=1}^{\infty} 4 \left(\frac{4}{3}\right)^{n-1}$$

with common ratio $r = \frac{4}{3} > 1$. By Theorem 6, the series is divergent.

□

Example 4.22. Find the sum of the series

$$\sum_{n=0}^{\infty} x^n, \quad \text{where } |x| < 1.$$

Solution. Note that $x^0 = 1$. (We adopt the convention that $x^0 = 1$ even when $x = 0$.)

The series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + x^4 + \cdots$$

is a geometric series with $a = 1$ and $r = x$. Since $|r| = |x| < 1$, the series converges and

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

□

Example 4.23. (Telescoping sums) Show that the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

is convergent, and find its sum.

Solution. The n -th term of the series can be written as (partial fraction decomposition)

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Thus, the partial sum is

$$\begin{aligned}
 s_k &= \sum_{n=1}^k \frac{1}{n(n+1)} \\
 &= \sum_{n=1}^k \left(\frac{1}{n} - \frac{1}{n+1} \right) \\
 &= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots + \left(\frac{1}{k} - \frac{1}{k+1} \right) \\
 &= 1 - \frac{1}{k+1}.
 \end{aligned}$$

Note that the terms in the second last equation cancel in pairs. This is an example of a **telescoping sum**.

Therefore,

$$\lim_{k \rightarrow \infty} s_k = \lim_{k \rightarrow \infty} \left(1 - \frac{1}{k+1} \right) = 1 - 0 = 1.$$

Hence, the given series converges, and the sum is

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1.$$

□

Example 4.24. Show that the **harmonic series**

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

is divergent.

Solution. We will show that the sequence of partial sums $\{s_k\}$ diverges to ∞ .

First, we calculate a lower bound for s_k , where k is a power of 2, as follows:

* If subsequence diverges, the sequence diverges.

$s_k = \frac{1}{2}$ for any k
 $\downarrow$
 hard to write

$$\begin{aligned}
s_2 &= 1 + \frac{1}{2}. \\
s_4 &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \\
&= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) \\
&> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right), \\
&> 1 + \frac{1}{2} + \frac{1}{2} = 1 + \frac{2}{2}.
\end{aligned}$$

$$\begin{aligned}
s_8 &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \\
&= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) \\
&> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) \\
&= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 1 + \frac{3}{2}.
\end{aligned}$$

$$\begin{aligned}
s_{16} &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \cdots + \frac{1}{16} \\
&= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \cdots + \frac{1}{16}\right) \\
&> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \cdots + \frac{1}{8}\right) + \left(\frac{1}{16} + \cdots + \frac{1}{16}\right) \\
&= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 1 + \frac{4}{2}.
\end{aligned}$$

In general, we have

$$s_{2^m} > 1 + \frac{m}{2}.$$

This shows that $s_{2^m} \rightarrow \infty$ as $m \rightarrow \infty$. By the Subsequence Test (Theorem 1), we deduce that s_k diverges, i.e. the harmonic series diverges. $\square$

p series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ $\rightarrow$ converges if $p > 1$
 $\rightarrow$ diverges if $p \leq 1$

Theorem 7. If the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$.

Proof. Let $s_k = a_1 + a_2 + \cdots + a_k$. Then

$$a_k = s_k - s_{k-1}. \quad (4.5)$$

Since $\sum a_n$ is convergent, we have

$$\lim_{k \rightarrow \infty} s_k = S$$

for some real number S . Also, clearly,

$$\lim_{k \rightarrow \infty} s_{k-1} = \lim_{k \rightarrow \infty} s_k = S.$$

It follows from (4.5) that

$$\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} (s_k - s_{k-1}) = \lim_{k \rightarrow \infty} s_k - \lim_{k \rightarrow \infty} s_{k-1} = S - S = 0.$$

□

NOTE: The converse of Theorem 7 is **NOT** true: If $a_n \rightarrow 0$ as $n \rightarrow \infty$, we cannot conclude that $\sum a_n$ is convergent. For example, $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$, but the series $\sum \frac{1}{n}$ is not convergent since this is the harmonic series. *→ harmonic series diverges*

The contrapositive of the statement of Theorem 7 can be used to test for divergence.

Theorem 8. (The n th Term Test for Divergence)

If $\lim_{n \rightarrow \infty} a_n$ does not exist or if $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

Example 4.25. Show that the series $\sum_{n=1}^{\infty} \frac{n^2}{5n^2 + 4}$ diverges.

Solution. Let $a_n = \frac{n^2}{5n^2 + 4}$. Then

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \frac{n^2}{5n^2 + 4} \\ &= \lim_{n \rightarrow \infty} \frac{1}{5 + 4/n^2} \\ &= \frac{1}{5} \neq 0. \end{aligned}$$

So the series diverges by the n th Term Test for Divergence. $\square$

NOTE: If we find $\lim_{n \rightarrow \infty} a_n \neq 0$, then we know that $\sum a_n$ diverges by the n th Term Test for Divergence. However, if we find $\lim_{n \rightarrow \infty} a_n = 0$, we know nothing about the convergence or divergence of $\sum a_n$.

Convergent series can be ‘added to’ and ‘subtracted from’ each other. The following properties follow from the corresponding Limit Laws for sequence.

Theorem 9. If $\sum a_n$ and $\sum b_n$ are convergent series, then

(i) $\sum ca_n$ is convergent (where c is some constant) and is equal to

$$\sum ca_n = c \sum a_n.$$

(ii) $\sum (a_n + b_n)$ is convergent, and is equal to

$$\sum (a_n + b_n) = \sum a_n + \sum b_n.$$

(iii) $\sum (a_n - b_n)$ is convergent, and is equal to

$$\sum (a_n - b_n) = \sum a_n - \sum b_n.$$

Example 4.26. Find the sum of the series

$$\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right).$$

Solution. The series $\sum_{n=1}^{\infty} \frac{1}{2^n}$ is a geometric series with $a = \frac{1}{2}$ and $r = \frac{1}{2}$, so

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{a}{1-r} = \frac{1/2}{1-1/2} = 1.$$

By Example 4.23, the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges to 1, so

$$\sum_{n=1}^{\infty} \frac{3}{n(n+1)} = 3 \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 3 \cdot 1 = 3.$$

Hence,

$$\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right) = \sum_{n=1}^{\infty} \frac{3}{n(n+1)} + \sum_{n=1}^{\infty} \frac{1}{2^n} = 3 + 1 = 4.$$

□

NOTE: A finite number of terms does not affect the convergence or divergence of a series.

Suppose we were able to show that the series

$$\sum_{n=4}^{\infty} \frac{n}{n^3 + 1}$$

is convergent. Then by adding a finite number of terms will not change the convergence. For example,

$$\sum_{n=1}^{\infty} \frac{n}{n^3 + 1} = \frac{1}{2} + \frac{2}{9} + \frac{3}{28} + \sum_{n=4}^{\infty} \frac{n}{n^3 + 1}$$

is still convergent.